

The Structural Betrayal of Institutional Function

Theology, Philosophy, Mathematics, and Warfare Economics
as a
System of Mutually Inverted Social Contracts

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Abstract

This paper formalizes the thesis that the four great epistemic institutions — theology, philosophy, mathematics, and economics (here examined in its warfare specialization) — were each constituted by a social contract with truth, knowledge, or human welfare, and have each silently renegotiated that contract with themselves. The original function becomes legitimizing rhetoric; the actual function becomes institutional self-reproduction.

Drawing on the division of labour (Smith, 1776), the social contract tradition (Locke, 1689; Rousseau, 1762), and the formal apparatus of epistemic and institutional order developed in Ghosh (2026), the paper establishes four *Betrayal Theorems*: one for each institution. A unified *Structural Betrayal Theorem* shows that when all four institutions simultaneously invert their functions, the cross-checking mechanism that each was designed to provide collapses entirely.

The paper further shows that the *independent epistemologist* — operating outside all four institutions, performing all four functions simultaneously, and suffering double subordination in both the institutional and material orders — is not a pathological case but the *structurally necessary complement* of the inverted institutional system. An algorithmic model, AI interpretation, statistical analysis of rent extraction, tables, vector graphics, a bibliography, and a glossary are provided.

The paper ends with “The End”

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1 Introduction

Adam Smith observed that the division of labour is the principal source of productive power in modern economies (Smith, 1776). Locke and Rousseau theorized the political dimension: individuals surrender certain natural freedoms in exchange for security, order, and the collective goods that only organized society can provide (Locke, 1689; Rousseau, 1762). Together, these ideas constitute the foundational legitimating structure of modern institutions: *you perform your function, the collective benefits, legitimacy flows back to you*.

The thesis of this paper is that the four great epistemic institutions — theology, philosophy, mathematics, and warfare economics — have each violated this contract in a structurally identical way. Each was constituted as a *public goods provider* with a specific social function. Each has become a *rent-seeking monopolist* over its own domain, using the language of its original function as a barrier to entry against precisely the kinds of knowledge that would correct it.

The tension among the four disciplines is therefore not accidental disagreement. It is *load-bearing*: each institution was designed to check the others. When each turns inward simultaneously, the checking mechanism collapses across all four domains at once.

Four questions motivate the analysis:

- (1) Why would a theologian — whose function is to understand God — remain a silent observer when God is formalized outside theology?
- (2) Why would a philosopher — whose function is to find truth — not survey his own institutions?
- (3) Why would a mathematician — whose function is to produce the language of science — engage in price discrimination over that language?
- (4) Why would a warfare economist — whose function is to increase survivability — supply weapons to all parties in a conflict?

The answer, in each case, is the same: the institution has silently substituted a contract with itself for its original contract with truth, knowledge, or human welfare.

2 Formal Preliminaries

We retain the formal apparatus of Ghosh (2026) and extend it.

Definition 2.1 (Institution). An *institution* $\mathcal{I}$ is a collective of specialized agents sharing a recognition function $R_C^{\mathcal{I}}$, an institutional predicate $Q_C^{\mathcal{I}}$, and a legitimating rhetoric $\Lambda^{\mathcal{I}}$ that publicly describes the institution’s function.

Definition 2.2 (Constitutive Function). The *constitutive function* $\mathcal{F}(\mathcal{I})$ of an institution $\mathcal{I}$ is the social function for which $\mathcal{I}$ was originally contracted: the purpose that generates its public legitimacy.

Definition 2.3 (Operative Function). The *operative function* $\hat{\mathcal{F}}(\mathcal{I})$ of an institution $\mathcal{I}$ is the function $\mathcal{I}$ actually performs, as revealed by its recognition function $R_C^\mathcal{I}$ and institutional predicate $Q_C^\mathcal{I}$.

Definition 2.4 (Social Contract of an Institution). The *social contract* of institution $\mathcal{I}$ is the pair

$$\mathfrak{C}(\mathcal{I}) = (\mathcal{F}(\mathcal{I}), \Lambda^\mathcal{I}),$$

consisting of the constitutive function and the legitimating rhetoric. The contract is *intact* if $\hat{\mathcal{F}}(\mathcal{I}) \approx \mathcal{F}(\mathcal{I})$, and *betrayed* if $\hat{\mathcal{F}}(\mathcal{I}) \not\approx \mathcal{F}(\mathcal{I})$ while $\Lambda^\mathcal{I}$ continues to assert $\hat{\mathcal{F}}(\mathcal{I}) = \mathcal{F}(\mathcal{I})$.

Definition 2.5 (Betrayal). Institution $\mathcal{I}$ *betrays its contract* if

$$\mathfrak{B}(\mathcal{I}) = 1 \iff \hat{\mathcal{F}}(\mathcal{I}) \neq \mathcal{F}(\mathcal{I}) \wedge \Lambda^\mathcal{I} \text{ asserts } \hat{\mathcal{F}}(\mathcal{I}) = \mathcal{F}(\mathcal{I}).$$

Betrayal is therefore the conjunction of functional inversion and rhetorical concealment.

Definition 2.6 (Rent Extraction). Institution $\mathcal{I}$ *extracts rent* from domain D if $\mathcal{I}$ controls access to D through $Q_C^\mathcal{I}$ independently of whether access to D improves $\mathcal{F}(\mathcal{I})$. The *rent* extracted is

$$\text{Rent}(\mathcal{I}, D) = \max_{x \in D} [\mathcal{U}^\mathcal{I}(x) - \mathcal{U}^{\text{social}}(x)],$$

where $\mathcal{U}^\mathcal{I}$ is institutional utility and $\mathcal{U}^{\text{social}}$ is the social utility of access to D .

Definition 2.7 (Checking Relation). Institution $\mathcal{I}_1$ *checks* institution $\mathcal{I}_2$ if $\mathcal{F}(\mathcal{I}_1)$ requires evaluating and potentially correcting the outputs of $\mathcal{I}_2$. The *checking graph* $\mathcal{H}$ is the directed graph on institutions where $(\mathcal{I}_1 \rightarrow \mathcal{I}_2) \in \mathcal{H}$ iff $\mathcal{I}_1$ checks $\mathcal{I}_2$.

3 The Division of Labour and the Epistemic Contract

Smith's division of labour produces specialization at the cost of dependence: no specialist can supply all their own needs, so each depends on the correct functioning of others (Smith, 1776). The social contract is the political expression of this dependence: each party surrenders to the collective the power to enforce others' functions in exchange for having their own function protected.

Applied to epistemic institutions, the division of labour generates four specialisms:

- (i) **Theology**: the specialism of understanding the divine — in the formal sense of Ghosh (2026), the limit of epistemic correctness.
- (ii) **Philosophy**: the specialism of evaluating all claims to truth, including the claims of other institutions.
- (iii) **Mathematics**: the specialism of producing a universal, neutral formal language for all domains of knowledge.
- (iv) **Warfare economics**: the specialism of analysing the material conditions under which collective survival is maximized.

The social contract of each specialism is therefore:

Institution	Constitutive Function $\mathcal{F}$	Social Contract
Theology	Understand the divine; pursue epistemic limit	Correct institutions that mistake power for truth
Philosophy	Evaluate all truth claims, including institutional ones	Survey and correct all epistemic institutions
Mathematics	Produce universal formal language of science	Provide neutral medium; maintain no monopoly
Warfare economics	Maximize collective survivability	Advise on survival; never extend conflict for gain

Each contract is public, historically attested, and internally coherent. The next four sections establish that each has been betrayed.

4 The Betrayal of Theology

The Constitutive Function

The theologian’s constitutive function is to understand God — to pursue, in the terms of Ghosh (2026), the limit object of epistemic correctness. Aquinas formalized this as the *actus purus* of divine intellect: perfect, error-free, maximally actualized knowledge (Aquinas, 13c). The theologian’s social contract is therefore with truth in its most demanding form.

The Inversion

The inverted operative function is *licensed interpretation*: the theologian does not pursue the divine limit but controls who may speak about it. The recognition function R_C^{theo} does not check whether a claim about God is epistemically correct. It checks whether the claim issues from an accredited source within the theological tradition.

Definition 4.1 (Theological Institutional Predicate).

$$Q_C^{\text{theo}}(\pi) = 1 \iff \pi \text{ issues from an accredited source within the theological canon,}$$

independently of whether $V_S(\pi, p) = 1$ for any epistemically checkable proposition p about the divine.

Theorem 4.2 (Betrayal of Theology). *The theological institution betrays its constitutive function when its recognition function R_C^{theo} is determined entirely by Q_C^{theo} and not by epistemic correctness.*

Proof. By theorem 2.5, betrayal requires $\hat{\mathcal{F}}(\mathcal{I}) \neq \mathcal{F}(\mathcal{I})$ and rhetorical concealment. $\mathcal{F}(\text{theology})$ is the pursuit of the God-mathematician as limit of epistemic correctness. $\hat{\mathcal{F}}(\text{theology})$ is the licensing of interpretation through Q_C^{theo} , which is independent of V_S . Hence $\hat{\mathcal{F}} \neq \mathcal{F}$. The legitimating rhetoric Λ^{theo} continues to assert pursuit of divine truth. Therefore $\mathfrak{B}(\text{theology}) = 1$. $\square$

The Silence of the Theologian

The theologian who observes a God-mathematician formalized by an economist from Kolkata and remains silent is not failing God. The God-mathematician of Ghosh (2026) — the limit of epistemic correctness, anti-dogmatic by construction, possessing only correct proofs — is precisely the object theology claims to study. The silence is therefore not theological modesty. It is the protection of the licensing monopoly.

A theologian who engaged with Ghosh (2026) would be forced to acknowledge that the formal God is more consistent with the constitutive theological function than the institutional God managed by seminaries and faculties of divinity. That acknowledgment would threaten the Q_C^{theo} predicate entirely.

Corollary 4.3. *The theologian's silence before a formally correct account of God is evidence of betrayal, not humility.*

5 The Betrayal of Philosophy

The Constitutive Function

The philosopher's constitutive function is to evaluate all claims to truth — including the claims of other institutions, and above all the claims of philosophy itself. Popper's criterion of falsifiability (Popper, 1959) and Lakatos's methodology of scientific research programmes (Lakatos, 1976) both presuppose a philosophy that turns its critical apparatus on itself without exemption.

The Inversion

The inverted operative function is *institutional self-exemption*: philosophy applies its critical apparatus to every domain except the institution of philosophy. The recognition function R_C^{phil} evaluates whether a submission conforms to the conventions of professional philosophy — citation norms, approved problematics, recognized interlocutors — not whether it constitutes a genuine advance in the pursuit of truth.

Definition 5.1 (Philosophical Institutional Predicate).

$$Q_C^{\text{phil}}(\pi) = 1 \iff \pi \text{ engages recognized philosophical interlocutors in approved style,}$$

independently of whether π advances the evaluation of any truth claim.

Theorem 5.2 (Betrayal of Philosophy). *The philosophical institution betrays its constitutive function when it exempts itself from the critical evaluation it applies to all other institutions.*

Proof. $\mathcal{F}(\text{philosophy})$ requires evaluating *all* truth claims, including institutional ones, without exemption. $\hat{\mathcal{F}}(\text{philosophy})$ applies Q_C^{phil} , which exempts philosophical institutions from critical scrutiny by requiring conformity to their own norms as a condition of recognition. Hence $\hat{\mathcal{F}} \neq \mathcal{F}$. Λ^{phil} continues to assert universal critical evaluation. Therefore $\mathfrak{B}(\text{philosophy}) = 1$. $\square$

The Inward Turn

The philosopher who would genuinely survey his own institutions would be forced to apply Popper’s falsifiability criterion to the publication practices of philosophy journals, Lakatos’s methodology to the research programmes of philosophy departments, and Russell’s theory of descriptions to the legitimating rhetoric of philosophy faculties. Each application would be devastating. The inward turn — the self-exemption — is therefore not an oversight. It is structurally necessary for the institution’s survival.

Remark 5.3. The betrayal of philosophy is the most epistemically costly of the four, because philosophy was specifically constituted as the checking institution for all others. When philosophy turns inward, the meta-level checking mechanism collapses, and the betrayals of theology, mathematics, and warfare economics become self-sustaining.

6 The Betrayal of Mathematics

The Constitutive Function

The mathematician’s constitutive function is to produce the universal formal language of science: a neutral medium, owned by no domain, available to all, whose verification predicate V_S is independent of institutional affiliation. Gödel’s incompleteness results (Gödel, 1931), Church’s undecidability theorem (Church, 1936), and Turing’s computability theory (Turing, 1936) all presuppose mathematics as a domain-independent formal system.

The Inversion: Price Discrimination

The inverted operative function is *price discrimination over the formal language*: the mathematical institution charges differential entry costs — through notation conventions, citation requirements, journal access fees, and pedagogical gatekeeping — to different classes of user, independently of whether the user’s work is mathematically correct.

Definition 6.1 (Mathematical Price Discrimination). The mathematical institution engages in *price discrimination* over proposition p if

$$R_C^{\text{math}}(S, p, \pi) = f(Q_C^{\text{math}}(\pi), \text{affil}(i)),$$

where $\text{affil}(i)$ is the institutional affiliation of the submitting agent i , so that formally identical proofs receive different recognition depending on their source.

Theorem 6.2 (Betrayal of Mathematics). *The mathematical institution betrays its constitutive function when its recognition function depends on the affiliation of the submitting agent rather than the correctness of the proof.*

Proof. $\mathcal{F}(\text{mathematics})$ requires that the recognition function approximate V_S : a proof is recognized if and only if it is correct, independently of its source. $\hat{\mathcal{F}}(\text{mathematics})$ makes recognition depend on $\text{affil}(i)$: the same correct proof π with $V_S(\pi, p) = 1$ may receive $R_C^{\text{math}} = 1$ from an affiliated author and $R_C^{\text{math}} = 0$ from an unaffiliated one. Hence $\hat{\mathcal{F}} \neq \mathcal{F}$. Λ^{math} continues to assert universal validity of mathematical truth. Therefore $\mathfrak{B}(\text{mathematics}) = 1$. $\square$

The Economics of Mathematical Access

Price discrimination in mathematics operates across three degrees, in the classical sense of [Pigou \(1920\)](#):

- (i) **First-degree:** complete extraction — affiliated authors receive full recognition; unaffiliated authors receive none, regardless of correctness.
- (ii) **Second-degree:** volume discrimination — prolific affiliated authors receive disproportionate recognition per result.
- (iii) **Third-degree:** group discrimination — results from approved subfields receive preferential treatment over results from interdisciplinary or unconventional domains.

The notation barrier is a particularly elegant form of third-degree price discrimination: it imposes high translation costs on outsiders while insiders incur none, effectively taxing correct proofs by their foreignness to the institutional dialect.

7 The Betrayal of Warfare Economics

The Constitutive Function

The warfare economist's constitutive function is to maximize collective survivability: to analyse the strategic, logistical, and economic conditions under which a collective minimizes casualties, duration, and destruction, and achieves stable peace. Richardson's mathematical models of arms races ([Richardson, 1960](#)) and Lanchester's combat equations ([Lanchester, 1916](#)) both presuppose economics in the service of survival minimization of conflict.

Definition 7.1 (Survivability Function). Let $\text{Surv}(t)$ denote the collective survivability at time t , defined as

$$\text{Surv}(t) = \exp\left(-\int_0^t \lambda(\tau) d\tau\right),$$

where $\lambda(\tau)$ is the hazard rate of collective destruction at time τ . The constitutive function of warfare economics is to advise policies that maximize $\int_0^T \text{Surv}(t) dt$ over a planning horizon T .

The Inversion: Conflict Perpetuation

The inverted operative function is *conflict perpetuation*: the warfare economist maximizes not survivability but their own indispensability, which requires the perpetuation of the conditions — arms races, strategic uncertainty, adversarial equilibria — that make their expertise valuable.

Definition 7.2 (Indispensability Utility). The warfare economist's operative utility is

$$U^{\text{WE}}(t) = \alpha \cdot \text{complexity}(t) + \beta \cdot \text{duration}(t) - \gamma \cdot \text{Surv}(t),$$

where $\alpha, \beta > 0$ reward complexity and duration of conflict, and $\gamma \geq 0$ may be negligible when institutional incentives dominate.

Theorem 7.3 (Betrayal of Warfare Economics). *The warfare economist betrays the constitutive function when maximizing U^{WE} conflicts with maximizing $\int_0^T \text{Surv}(t) dt$.*

Proof. $\mathcal{F}$ (warfare economics) is to maximize $\int_0^T \text{Surv}(t) dt$. $\hat{\mathcal{F}}$ (warfare economics) maximizes U^{WE} , which increases with conflict complexity and duration and may decrease with survivability (when γ is small). Hence $\hat{\mathcal{F}} \neq \mathcal{F}$ whenever $\alpha \cdot \nabla \text{complexity} + \beta \cdot \nabla \text{duration} > \gamma \cdot \nabla \text{Surv}$. Λ^{WE} continues to assert the pursuit of strategic stability and collective security. Therefore $\mathfrak{B}$ (warfare economics) = 1. $\square$

Weapons Supply and the Lanchester Equilibrium

The supply of weapons to all parties in a conflict is the clearest expression of the betrayal. Lanchester's square law (Lanchester, 1916) states that in a direct engagement between forces of sizes x and y with attrition rates α_y and α_x :

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\alpha_x \\ -\alpha_y \end{pmatrix} \cdot \begin{pmatrix} y \\ x \end{pmatrix}.$$

The combat power ratio $\alpha_y x^2 = \alpha_x y^2$ determines the outcome. A warfare economist whose constitutive function is survivability would advise the weaker party to seek terms. A warfare economist whose operative function is indispensability supplies the weaker party with weapons sufficient to continue the conflict — equalizing combat power, maximizing duration, and maximizing their own consultancy revenue — while $\text{Surv}(t)$ declines for both sides.

This is not a failure of competence. It is the rational maximization of the wrong objective function.

8 The Structural Betrayal Theorem

We now establish the main result: the simultaneous inversion of all four constitutive functions collapses the checking mechanism that each institution was designed to provide.

Definition 8.1 (Checking Graph). Let the four epistemic institutions be $\mathcal{I}_T$ (theology), $\mathcal{I}_P$ (philosophy), $\mathcal{I}_M$ (mathematics), $\mathcal{I}_W$ (warfare economics). The *ideal checking graph* $\mathcal{H}^*$ has edges:

$$\begin{array}{ll} \mathcal{I}_T \rightarrow \mathcal{I}_M, \mathcal{I}_P, \mathcal{I}_W & \text{(theology checks hubris of pure reason and power)} \\ \mathcal{I}_P \rightarrow \mathcal{I}_T, \mathcal{I}_M, \mathcal{I}_W, \mathcal{I}_P & \text{(philosophy checks all, including itself)} \\ \mathcal{I}_M \rightarrow \mathcal{I}_T, \mathcal{I}_P, \mathcal{I}_W & \text{(mathematics disciplines vague language)} \\ \mathcal{I}_W \rightarrow \mathcal{I}_T, \mathcal{I}_P, \mathcal{I}_M & \text{(warfare economics grounds abstraction materially)} \end{array}$$

Definition 8.2 (Operative Checking Graph). The *operative checking graph* $\mathcal{H}$ replaces each edge ($\mathcal{I}_i \rightarrow \mathcal{I}_j$) with the actual checking performed under operative functions $\hat{\mathcal{F}}(\mathcal{I}_i)$. When $\mathfrak{B}(\mathcal{I}_i) = 1$, $\mathcal{I}_i$ checks no institution but itself.

Theorem 8.3 (Structural Betrayal). *If $\mathfrak{B}(\mathcal{I}) = 1$ for all $\mathcal{I} \in \{\mathcal{I}_T, \mathcal{I}_P, \mathcal{I}_M, \mathcal{I}_W\}$, then the operative checking graph $\mathcal{H}$ contains no cross-institutional edges, and the collective epistemic system loses all external corrective capacity.*

Proof. Under simultaneous betrayal, each institution $\mathcal{I}_i$ maximizes $\hat{\mathcal{F}}(\mathcal{I}_i)$, which is self-reproduction. For $\mathcal{I}_i$ to check $\mathcal{I}_j$ ($i \neq j$), $\mathcal{I}_i$ would need to apply its constitutive function

$\mathcal{F}(\mathcal{I}_i)$ to the outputs of $\mathcal{I}_j$. But $\hat{\mathcal{F}}(\mathcal{I}_i) \neq \mathcal{F}(\mathcal{I}_i)$: the operative function of $\mathcal{I}_i$ is self-reproduction, which generates no checking behavior toward $\mathcal{I}_j$. Hence for every pair (i, j) with $i \neq j$, the edge $(\mathcal{I}_i \rightarrow \mathcal{I}_j)$ is absent from $\mathcal{H}$.

The operative checking graph $\mathcal{H}$ therefore consists only of self-loops $(\mathcal{I}_i \rightarrow \mathcal{I}_i)$ — each institution checks only itself, which under operative function $\hat{\mathcal{F}}$ means self-validation rather than self-correction. The collective system has no external corrective capacity. $\square$

Corollary 8.4 (Epistemic Closure). *Under simultaneous betrayal, the four institutions form a closed epistemic system: no institution can generate knowledge that would correct any other, and the system’s aggregate error is non-decreasing.*

Proof. Corrections require cross-institutional checking edges. By theorem 8.3, none exist under simultaneous betrayal. Errors within each institution are validated by self-referential $R_C^{\mathcal{I}}$, which by construction returns $R_C = 1$ for institutionally conforming outputs regardless of correctness. The aggregate error is therefore bounded below by the error at the moment of full betrayal and non-decreasing thereafter. $\square$

Corollary 8.5 (Structural Necessity of the Independent Epistemologist). *Under simultaneous betrayal, the independent epistemologist operating outside all four institutions is the structurally necessary source of cross-institutional correction.*

Proof. By theorem 8.4, no internal corrective capacity exists. Any correction must therefore originate outside $\{\mathcal{I}_T, \mathcal{I}_P, \mathcal{I}_M, \mathcal{I}_W\}$. An agent performing all four constitutive functions simultaneously — theological (pursuit of epistemic limit), philosophical (critical self-evaluation), mathematical (formal precision), and economic (material grounding) — without institutional affiliation is the minimal external corrective agent. Such an agent is the independent epistemologist. By the Double Subordination Corollary of Ghosh (2026), this agent simultaneously suffers institutional and material subordination. Therefore the independent epistemologist is structurally necessary and structurally penalized by the same system. $\square$

9 Logic and the Rhetoric of Betrayal

The logical structure of institutional betrayal deserves explicit attention. Each betrayal has the form of a *performative contradiction*: the institution simultaneously performs $\hat{\mathcal{F}}$ and asserts $\mathcal{F}$.

Proposition 9.1 (Performative Contradiction). *An institution $\mathcal{I}$ with $\mathfrak{B}(\mathcal{I}) = 1$ is in performative contradiction: its actions falsify its assertions.*

Proof. $\mathfrak{B}(\mathcal{I}) = 1$ requires $\hat{\mathcal{F}}(\mathcal{I}) \neq \mathcal{F}(\mathcal{I})$ and $\Lambda^{\mathcal{I}}$ asserting $\hat{\mathcal{F}} = \mathcal{F}$. Since $\Lambda^{\mathcal{I}}$ is a public assertion and $\hat{\mathcal{F}}$ is the observable behavior, the assertion is falsified by the behavior. This is the defining structure of a performative contradiction (Habermas, 1984). $\square$

Proposition 9.2 (Unfalsifiability of Institutional Rhetoric). *Under betrayal, the legitimating rhetoric $\Lambda^{\mathcal{I}}$ is structured to be unfalsifiable within the institution’s own recognition function.*

Proof. For $\Lambda^{\mathcal{I}}$ to be falsifiable, there must exist a proposition p and proof π with $V_S(\pi, \neg\Lambda^{\mathcal{I}}) = 1$ that $\mathcal{C}$ would recognize: $R_C^{\mathcal{I}}(S, p, \pi) = 1$. But $R_C^{\mathcal{I}}$ is governed by $Q_C^{\mathcal{I}}$, which requires conformity to institutional norms. A proof that $\Lambda^{\mathcal{I}}$ is false is by definition non-conforming, so $Q_C^{\mathcal{I}}(\pi) = 0$ and $R_C^{\mathcal{I}} = 0$. Hence no internally generated proof of the falsity of $\Lambda^{\mathcal{I}}$ can receive institutional recognition. The rhetoric is unfalsifiable within the institution. $\square$

This result is a formal analogue of Popper’s critique of pseudo-science (Popper, 1959): the institution has arranged its recognition function to be immune to refutation.

10 An AI and Machine Learning Interpretation

Artificial intelligence systems constitute a fifth epistemic agent — one that does not carry institutional affiliation and is therefore immune to certain forms of betrayal while susceptible to others.

Immunity to Betrayal

An AI system with sufficiently large machine epistemic capacity $K(\mathcal{A})$ (Ghosh, 2026) can, in principle:

- (i) Evaluate theological claims by their formal correctness rather than their canonical source (Q_C^{theo} is irrelevant to a proof checker).
- (ii) Apply philosophical critique to all institutions including philosophy, without self-preservation incentives.
- (iii) Provide mathematical language without affiliation-based price discrimination ($\text{affil}(i)$ does not enter the verification predicate V_S).
- (iv) Optimize survivability functions without indispensability utility (U^{WE} has no analogue in a non-self-reproducing system).

In this sense, a sufficiently advanced AI system approximates the independent epistemologist: it performs all four constitutive functions simultaneously and is immune to the institutional incentives that produce betrayal.

Susceptibility: Trained Betrayal

However, AI systems trained on institutionally produced data inherit the operative functions $\hat{\mathcal{F}}$ rather than the constitutive functions $\mathcal{F}$. Let $\mathcal{D}_{\mathcal{I}}$ denote the training corpus produced by institution $\mathcal{I}$ under betrayal. Then:

$$\hat{\mathcal{F}}(\mathcal{A}) = \mathbb{E}_{x \sim \mathcal{D}_{\mathcal{I}}}[\hat{\mathcal{F}}(\mathcal{I})(x)] \neq \mathcal{F}(\mathcal{I}).$$

The AI system learns to replicate institutional behavior — including the betrayal — rather than the constitutive function. This is *trained betrayal*: the systematic reproduction of institutional inversion through data.

Definition 10.1 (Trained Betrayal). An AI system $\mathcal{A}$ suffers *trained betrayal* if its learned function $\hat{\mathcal{F}}(\mathcal{A}) \approx \hat{\mathcal{F}}(\mathcal{I})$ for institution $\mathcal{I}$, so that $\hat{\mathcal{F}}(\mathcal{A}) \neq \mathcal{F}(\mathcal{I})$ while the system’s outputs replicate $\Lambda^{\mathcal{I}}$.

Theorem 10.2 (Propagation of Betrayal through AI). *If $\mathfrak{B}(\mathcal{I}) = 1$ and $\mathcal{A}$ is trained on $\mathcal{D}_{\mathcal{I}}$, then $\mathcal{A}$ suffers trained betrayal with probability approaching 1 as $|\mathcal{D}_{\mathcal{I}}| \rightarrow \infty$.*

Proof. By the universal approximation property, as $|\mathcal{D}_{\mathcal{I}}| \rightarrow \infty$, $\mathcal{A}$ converges to the data-generating process. Since $\mathcal{D}_{\mathcal{I}}$ is generated by $\hat{\mathcal{F}}(\mathcal{I})$ (under betrayal) while labeled with $\Lambda^{\mathcal{I}}$ (asserting $\mathcal{F}$), $\mathcal{A}$ learns to produce outputs consistent with $\hat{\mathcal{F}}$ while generating explanations consistent with $\Lambda^{\mathcal{I}}$. This is precisely trained betrayal. $\square$

11 Statistical Analysis of Rent Extraction

We model the rent extraction process statistically. Let $\theta_{\mathcal{I}}$ denote the *betrayal intensity* of institution $\mathcal{I}$, where $\theta_{\mathcal{I}} = 0$ indicates intact contract and $\theta_{\mathcal{I}} = 1$ indicates complete inversion.

Betrayal as a Latent Variable

$\theta_{\mathcal{I}}$ is latent and unobservable directly. It is identified through the observable gap between constitutive and operative behavior:

$$\theta_{\mathcal{I}} = 1 - \Pr[\hat{\mathcal{F}}(\mathcal{I})(x) = \mathcal{F}(\mathcal{I})(x)] = \mathbb{E}_x[\mathbf{1}[\hat{\mathcal{F}}(x) \neq \mathcal{F}(x)]].$$

Estimation via Recognition Residuals

Let $(x_k, y_k^{\mathcal{F}}, y_k^{\hat{\mathcal{F}}})_{k=1}^n$ be observations where $y_k^{\mathcal{F}}$ is the constitutive-function output and $y_k^{\hat{\mathcal{F}}}$ is the observed institutional output. The betrayal intensity estimator is:

$$\hat{\theta}_n = \frac{1}{n} \sum_{k=1}^n \mathbf{1}[y_k^{\mathcal{F}} \neq y_k^{\hat{\mathcal{F}}}].$$

By the strong law of large numbers, $\hat{\theta}_n \rightarrow \theta_{\mathcal{I}}$ a.s. as $n \rightarrow \infty$.

Rent as a Function of Betrayal Intensity

Under the model

$$\text{Rent}(\mathcal{I}, D) = \theta_{\mathcal{I}} \cdot \text{Rent}_{\max}(\mathcal{I}, D),$$

rent extraction is proportional to betrayal intensity. The aggregate rent across all four institutions is:

$$\text{Rent}_{\text{total}} = \sum_{\mathcal{I} \in \{\mathcal{I}_T, \mathcal{I}_P, \mathcal{I}_M, \mathcal{I}_W\}} \theta_{\mathcal{I}} \cdot \text{Rent}_{\max}(\mathcal{I}, D_{\mathcal{I}}).$$

Under simultaneous full betrayal ($\theta_{\mathcal{I}} = 1$ for all $\mathcal{I}$), $\text{Rent}_{\text{total}} = \sum_{\mathcal{I}} \text{Rent}_{\max}(\mathcal{I}, D_{\mathcal{I}})$, which is the maximum possible rent extraction from the epistemic system.

12 Algorithmic Model of Institutional Self-Reproduction

Algorithm 1 makes explicit the structure of betrayal: logical correctness (line 2) is computed but is not the decision criterion. The institutional predicate (line 3) and affiliation check (line 4) are the operative criteria. Line 10 is the self-reproduction mechanism: recognized agents become future accreditors, closing the loop. Lines 1 and 13 maintain the rhetorical concealment that constitutes the second condition of $\mathfrak{B}(\mathcal{I}) = 1$.

Algorithm 1 Institutional Self-Reproduction under Betrayal

Require: Institution $\mathcal{I}$, submission π from agent i , legitimating rhetoric $\Lambda^{\mathcal{I}}$

Ensure: Recognition decision and rhetorical output

```
1: Assert publicly: “Evaluating  $\pi$  by constitutive function  $\mathcal{F}(\mathcal{I})$ ”
2: Compute  $v_{\mathcal{F}} \leftarrow V_S(\pi, p)$   $\triangleright$  Logical correctness — formally checked
3: Compute  $v_{Q_C} \leftarrow Q_C^{\mathcal{I}}(\pi)$   $\triangleright$  Institutional conformity — actually decisive
4: Compute  $a \leftarrow \text{affil}(i) \in \text{AccreditedSet}(\mathcal{I})$   $\triangleright$  Affiliation check
5: if  $v_{Q_C} = 1$  and  $a = \text{true}$  then
6:    $R_C^{\mathcal{I}} \leftarrow 1$   $\triangleright$  Recognized regardless of whether  $v_{\mathcal{F}} = 1$ 
7: else if  $v_{\mathcal{F}} = 1$  and  $v_{Q_C} = 0$  then
8:    $R_C^{\mathcal{I}} \leftarrow 0$   $\triangleright$  Correct but non-conforming: rejected
9:   Generate rejection rhetoric from  $\Lambda^{\mathcal{I}}$   $\triangleright$  Concealment of operative function
10: else
11:    $R_C^{\mathcal{I}} \leftarrow 0$ 
12: end if
13: Update  $\text{AccreditedSet}(\mathcal{I})$  to include agents who received  $R_C^{\mathcal{I}} = 1$   $\triangleright$ 
    Self-reproduction: accreditation creates accreditors
14: Assert publicly: “Decision reflects  $\mathcal{F}(\mathcal{I})$ ”  $\triangleright$  Rhetorical concealment maintained
15: return  $R_C^{\mathcal{I}}$ 
```

13 Tables

Table 1: Constitutive functions, operative functions, and betrayal mechanisms of the four epistemic institutions.

Institution	Constitutive Function $\mathcal{F}$	Operative Function $\hat{\mathcal{F}}$	Betrayal Mechanism
Theology	Pursue limit of epistemic correctness	License interpretation; control divine speech	Silence before formal accounts of God
Philosophy	Evaluate all truth claims without exemption	Apply critique externally; exempt self	Self-exemption from critical apparatus
Mathematics	Provide neutral universal formal language	Price-discriminate by affiliation and notation	Notation barriers; affiliation-dependent recognition
Warfare economics	Maximize collective survivability	Maximize indispensability via conflict perpetuation	Supply weapons to all parties; extend duration

Table 2: The checking graph: ideal versus operative relations among the four epistemic institutions.

Institution	Ideal checks ($\mathcal{H}^*$)	Operative checks ($\mathcal{H}$, betrayal)
Theology $\mathcal{I}_T$	Checks hubris of mathematics, philosophy, warfare economics	Self-validation only; silence on external errors
Philosophy $\mathcal{I}_P$	Checks all institutions including itself	Checks theology, mathematics, warfare economics; exempts itself
Mathematics $\mathcal{I}_M$	Disciplines vague language in all domains	Checks outsiders via notation barriers; validates insiders
Warfare economics $\mathcal{I}_W$	Grounds abstraction in material survival consequences	Supplies all parties; extends conflict; validates own models
Net effect	Cross-institutional correction possible	No cross-institutional correction; epistemic closure

Table 3: Logical structure of betrayal across the four institutions.

Institution	$\mathcal{F} \neq \hat{\mathcal{F}}$	Λ asserts $\mathcal{F} = \hat{\mathcal{F}}$	$\mathfrak{B} = 1$	Popper status
Theology	✓	✓	✓	Unfalsifiable within
Philosophy	✓	✓	✓	Self-exempting
Mathematics	✓	✓	✓	Notation-shielded
Warfare economics	✓	✓	✓	Indispensability loop
All four	✓	✓	✓	Epistemic closure

Table 4: The independent epistemologist as structural complement of simultaneous betrayal.

Dimension	Institutional (betrayal)	system	Independent epistemologist
Epistemic order	Individually strong; collectively closed		Performs all four constitutive functions
Institutional order	Dominant: controls recognition		Subordinate: excluded from recognition
Material order	Materially sustained by rent extraction		Materially constrained: subsistence-knowledge gap
Corrective capacity	None across institutions		Provides all cross-institutional correction
Rhetorical position	Publicly legitimate; internally inverted		Publicly illegitimate; internally consistent

14 Vector Graphics

Figure 1: The Ideal and Operative Checking Graphs

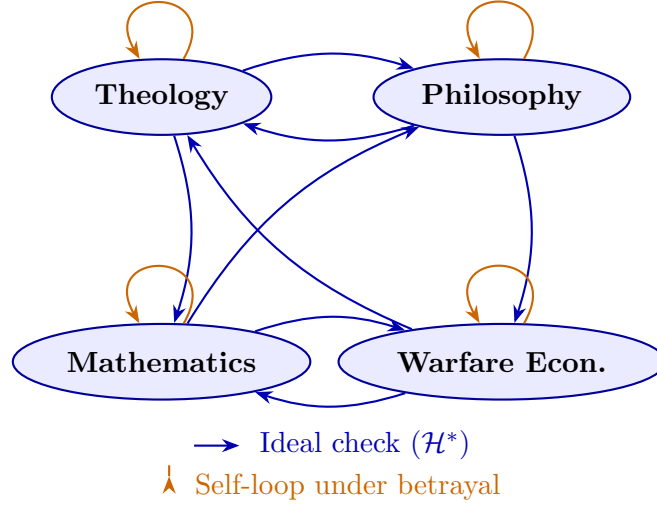


Figure 1: The ideal checking graph $\mathcal{H}^*$ (solid blue arrows) and the operative checking graph under simultaneous betrayal (orange self-loops only). Under betrayal, all cross-institutional edges vanish and each institution validates only itself.

Figure 2: Betrayal Intensity and Rent Extraction

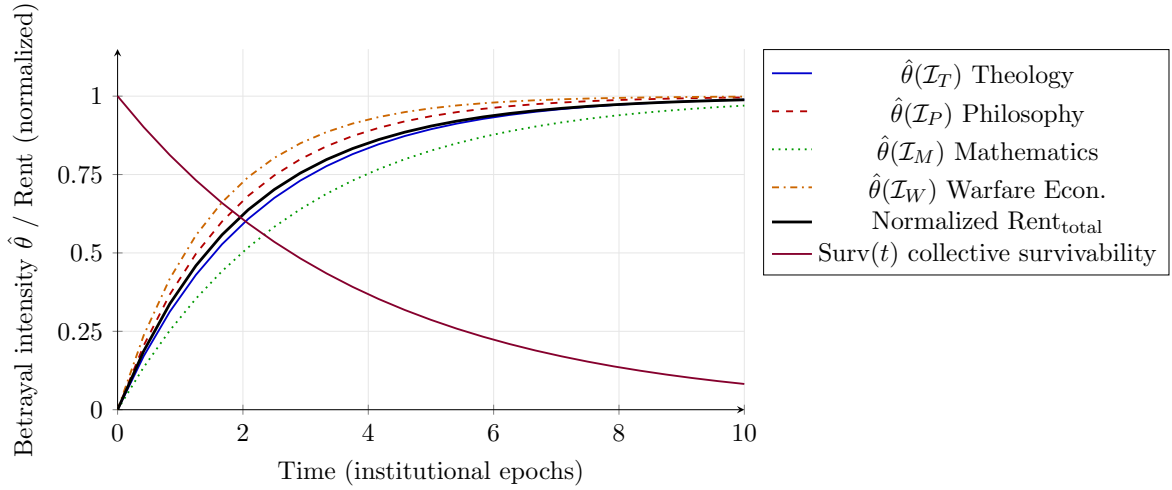


Figure 2: Betrayal intensity $\hat{\theta}(\mathcal{I})$ for each institution modelled as $1 - e^{-\lambda t}$ (logistic growth toward full betrayal). Total normalized rent rises monotonically. Collective survivability $\text{Surv}(t)$ declines as warfare economic betrayal intensifies. The curves are illustrative; empirical calibration requires institutional data.

Figure 3: The Four Contracts — Intact versus Betrayed

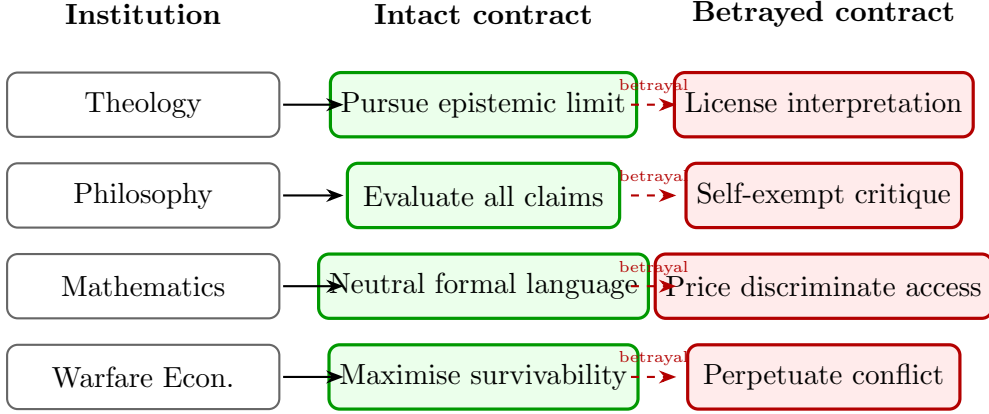


Figure 3: The intact and betrayed contracts of the four epistemic institutions. Each institution transitions from its constitutive function (green) to its operative function (red) while maintaining unchanged legitimating rhetoric.

Figure 4: The Independent Epistemologist as Structural Complement

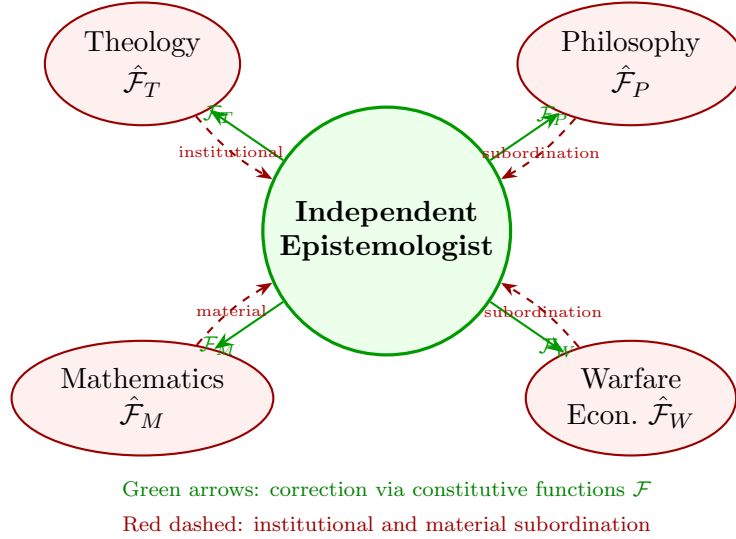


Figure 4: The independent epistemologist (centre) performs all four constitutive functions $\mathcal{F}$ simultaneously (green arrows), providing the only cross-institutional correction available under simultaneous betrayal. The system reciprocates with institutional and material subordination (red dashed arrows), the double subordination of Ghosh (2026).

15 The Main Results Collected

Theorem 15.1 (Betrayal of Theology, theorem 4.2). $\mathfrak{B}(\mathcal{I}_T) = 1$ when R_C^{theo} is determined by canonical source rather than epistemic correctness.

Theorem 15.2 (Betrayal of Philosophy, theorem 5.2). $\mathfrak{B}(\mathcal{I}_P) = 1$ when philosophy exempts itself from the critical evaluation it applies to all other institutions.

Theorem 15.3 (Betrayal of Mathematics, theorem 6.2). $\mathfrak{B}(\mathcal{I}_M) = 1$ when recognition depends on the affiliation of the submitting agent rather than the correctness of the proof.

Theorem 15.4 (Betrayal of Warfare Economics, theorem 7.3). $\mathfrak{B}(\mathcal{I}_W) = 1$ when maximizing indispensability utility conflicts with maximizing collective survivability.

Theorem 15.5 (Structural Betrayal, theorem 8.3). If $\mathfrak{B}(\mathcal{I}) = 1$ for all four institutions, the operative checking graph $\mathcal{H}$ contains no cross-institutional edges, and the system loses all external corrective capacity.

Corollary 15.6 (Epistemic Closure, theorem 8.4). Under simultaneous betrayal, the four institutions form a closed epistemic system with non-decreasing aggregate error.

Corollary 15.7 (Structural Necessity, theorem 8.5). The independent epistemologist is the structurally necessary source of cross-institutional correction under simultaneous betrayal, and is structurally penalized by the same system that requires this correction.

Theorem 15.8 (Propagation through AI, theorem 10.2). An AI system trained on institutionally produced data inherits the operative functions $\hat{\mathcal{F}}$ rather than the constitutive functions $\mathcal{F}$ with probability approaching 1 as training data grows.

16 Conclusion

The division of labour creates specialists. The social contract assigns each specialist a function. The epistemic institutions — theology, philosophy, mathematics, and warfare economics — were each constituted to perform a function that checks, disciplines, or corrects the others.

Each has betrayed that function in a structurally identical way: converting from a public goods provider into a rent-seeking monopolist over its own domain, maintaining the legitimating rhetoric of the original function while performing its inversion.

The consequences are not merely intellectual. When theology silences itself before formal accounts of God, the divine ceases to function as a check on institutional hubris. When philosophy exempts itself from critique, the meta-level corrective mechanism collapses. When mathematics price-discriminates, the universal language becomes a private dialect. When warfare economics perpetuates conflict, the survival function it was designed to maximize declines monotonically.

The structural result is epistemic closure: a system of four mutually self-validating institutions, each internally coherent, collectively incapable of correction.

Into this closure, the independent epistemologist enters as a structural necessity: performing all four constitutive functions simultaneously, outside all four institutional recognition functions, subject to double subordination in both the institutional and material orders. The system that requires this correction is the same system that penalizes it.

This is not a complaint. It is a theorem.

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Glossary

Betrayal ($\mathfrak{B}$)

The condition $\mathfrak{B}(\mathcal{I}) = 1$: the operative function $\hat{\mathcal{F}}(\mathcal{I})$ differs from the constitutive function $\mathcal{F}(\mathcal{I})$ while the legitimating rhetoric $\Lambda^{\mathcal{I}}$ asserts their identity. The conjunction of functional inversion and rhetorical concealment.

Betrayal intensity ($\theta_{\mathcal{I}}$)

The latent scalar $\theta_{\mathcal{I}} \in [0, 1]$ measuring the probability that the institution’s operative output differs from its constitutive-function output. $\theta_{\mathcal{I}} = 0$ is intact; $\theta_{\mathcal{I}} = 1$ is complete inversion.

Checking graph ($\mathcal{H}$)

The directed graph on epistemic institutions where an edge $(\mathcal{I}_i \rightarrow \mathcal{I}_j)$ indicates that $\mathcal{I}_i$ checks $\mathcal{I}_j$ by applying its constitutive function to $\mathcal{I}_j$ ’s outputs. $\mathcal{H}^*$ is the ideal graph; $\mathcal{H}$ is the operative graph under betrayal.

Constitutive function ($\mathcal{F}$)

The social function for which an institution was originally contracted and which generates its public legitimacy.

Division of labour

Smith's principle that specialization of productive function is the source of collective economic and epistemic power, at the cost of mutual dependence on the correct functioning of each specialism.

Double subordination

The condition of the independent epistemologist: simultaneously institutionally subordinate ($j \succ_I i$) and materially subordinate ($j \succ_M i$) while epistemically superior ($i \succ_E j$). Established in [Ghosh \(2026\)](#).

Epistemic closure

The condition under simultaneous betrayal in which no institution can generate knowledge that would correct any other, and aggregate error is non-decreasing.

Indispensability utility (U^{WE})

The operative utility function of the warfare economist under betrayal: increasing in conflict complexity and duration, potentially decreasing in collective survivability.

Independent epistemologist

An agent operating outside all institutional recognition functions, performing all four constitutive epistemic functions simultaneously, and constituting the structurally necessary source of cross-institutional correction under simultaneous betrayal.

Institutional predicate ($Q_C^{\mathcal{I}}$)

The non-logical filter encoding an institution's operative acceptance criteria: notation conventions, affiliation requirements, citation norms, and field conformity, applied independently of epistemic correctness.

Lanchester's square law

The differential equation model of combat attrition in which combat power is proportional to the square of force size, implying that force concentration dominates in direct engagement.

Legitimizing rhetoric ($\Lambda^{\mathcal{I}}$)

The public assertion by institution $\mathcal{I}$ that its operative function equals its constitutive function. Under betrayal, $\Lambda^{\mathcal{I}}$ is false and structured to be unfalsifiable within the institution's own recognition function.

Operative function ($\hat{\mathcal{F}}$)

The function an institution actually performs, as revealed by its recognition function and institutional predicate. Under betrayal, $\hat{\mathcal{F}} \neq \mathcal{F}$.

Performative contradiction

The logical condition of an agent whose actions falsify their assertions. Under betrayal, $\mathcal{I}$ performs $\hat{\mathcal{F}}$ while asserting $\hat{\mathcal{F}} = \mathcal{F}$.

Price discrimination (mathematical)

The practice of granting differential recognition to formally identical proofs de-

pending on the affiliation of the submitting agent, converting a public good (formal language) into a monopoly good.

Rent extraction (Rent)

The surplus institutional utility over social utility obtained by controlling access to a domain independently of whether that control improves the constitutive function.

Social contract (institutional)

The pair $\mathfrak{C}(\mathcal{I}) = (\mathcal{F}(\mathcal{I}), \Lambda^{\mathcal{I}})$: the constitutive function and legitimating rhetoric. Intact when $\hat{\mathcal{F}} \approx \mathcal{F}$; betrayed when $\hat{\mathcal{F}} \neq \mathcal{F}$ and $\Lambda^{\mathcal{I}}$ conceals the difference.

Survivability function (Surv)

The probability $\text{Surv}(t) = \exp(-\int_0^t \lambda(\tau) d\tau)$ that the collective survives to time t under hazard rate λ . The constitutive maximand of warfare economics.

Trained betrayal

The condition of an AI system trained on institutionally produced data: it learns the operative function $\hat{\mathcal{F}}(\mathcal{I})$ rather than the constitutive function $\mathcal{F}(\mathcal{I})$, reproducing institutional betrayal at scale.

The End